

Automotive Electronics Council

Component Technical Committee

Q100 Qualification Test Plan

Automotive Grade Level = {Select Grade}

MSL = {Select MSL}

Supplier Name:		General Specification:	AEC-Q100 Rev. H	
Supplier Code:		Supplier Wafer Fabrication:		
Supplier Part Number:		Supplier Wafer Test:		
Supplier Contact:		Supplier Assembly Site:		
Supplier Family Type:		Supplier Final Test Site:		
Device Description:		Supplier Reliability Signature:		
PPAP Submission Date:		Customer Test ID:		
Reason for Qualification:	{Select Reason}	Customer Part Number:		
Prepared by Signature:		Date:	Customer Approval Signature:	
				Date:

Test	#	Reference	Test Conditions	Lots	S.S.	Total	Results Lot/Pass/Fail	Comments: (N/A =Not Applicable)
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TEST GROUP A – ACCELERATED ENVIRONMENT STRESS TESTS

PC	A1	JESD22 A113 J-STD-020	Preconditioning: (Test @ Rm) SMD only; Moisture Preconditioning for THB/HAST, AC/UHST, TC, & PTC; Peak Reflow Temp =	Min. MSL = 3			MSL =	
THB or HAST	A2	JESD22 A101 JESD22 A110	Temperature Humidity Bias: (Test @ Rm/Hot) Highly Accelerated Stress Test: (Test @ Rm/Hot/)	3	77	231	of	
AC or UHST or TH	A3	JESD22 A102 JESD22 A118 or JESD22-A101	Autoclave: (Test @ Rm) Unbiased Highly Accelerated Stress Test: (Test @ Rm) Temperature Humidity without Bias: (Test @ Rm)	3	77	231	of	
TC	A4	JESD22 A104	Temperature Cycle: (Test @ Hot)	3	77	231	of	
PTC	A5	JESD22 A105	Power Temperature Cycle: (Test @ Rm/Hot)	1	45	45	of	
HTSL	A6	JESD22 A103	High Temperature Storage Life: (Test @ Rm/Hot)	1	45	45	of	

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TEST GROUP B – ACCELERATED LIFETIME SIMULATION TESTS

HTOL	B1	JESD22 A108	High Temp Operating Life: (Test @ Rm/Cold/Hot)	3	77	231	of	
ELFR	B2	AEC-Q100-008	Early Life Failure Rate: (Test @ Rm/Hot)	3	800	2400	of	
EDR	B3	AEC-Q100-005	NVM Endurance & Data Retention Test: (Test @ Rm/Hot)	3	77	231	of	

TEST GROUP C – PACKAGE ASSEMBLY INTEGRITY TESTS

WBS	C1	AEC-Q100-001 AEC-Q003	Wire Bond Shear Test: (Cpk > 1.67)	30 bonds	5 parts Min.	bonds	of	
WBP	C2	Mil-STD-883, Method 2011 AEC-Q003	Wire Bond Pull: (Cpk > 1.67); Each bonder used	30 bonds	5 parts Min.	bonds	of	
SD	C3	JESD22 B102 JSTD-002D	Solderability: (>95% coverage) 8hr steam aging prior to testing	1	15	15	of	
PD	C4	JESD22 B100, JESD22 B108 AEC-Q003	Physical Dimensions: (Cpk > 1.67)	3	10	30	of	
SBS	C5	AEC-Q100-010 AEC-Q003	Solder Ball Shear: (Cpk > 1.67); 5 balls from min. of 10 devices	3	50 balls		of	
LI	C6	JESD22 B105	Lead Integrity: (No lead cracking or breaking); Through- hole only; 10 leads from each of 5 devices	1	50 leads		of	

TEST GROUP D – DIE FABRICATION RELIABILITY TESTS

EM	D1	JESD61	Electromigration:	-	-	-		Data Available
Tddb	D2	JESD35	Time Dependant Dielectric Breakdown:	-	-	-		Data Available
HCI	D3	JESD60 & 28	Hot Carrier Injection:	-	-	-		Data Available

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NBTI	D4	JESD90	Negative Bias Temperature Instability:	-	-	-		Data Available
SM	D5	JESD61, 87, & 202	Stress Migration:	-	-	-		Data Available

TEST GROUP E- ELECTRICAL VERIFICATION

TEST	E1	User/Supplier Specification	Pre and Post Stress Electrical Test:	All	All	All	of	
HBM	E2	AEC-Q100-002	Electrostatic Discharge, Human Body Model: (Test @ Rm/Hot); (2KV HBM / Class 2 or better)	1			of ESD Level = {Select HBM Level}	
CDM	E3	AEC-Q100-011	Electrostatic Discharge, Charged Device Model: (Test @ Rm/Hot); (750V corner leads, 500V all other leads / Class C4B or better)	1			of ESD Level = {Select CDM Level}	
LU	E4	AEC-Q100-004	Latch-Up: (Test @ Rm/Hot)	1	6	6	of	
ED	E5	AEC-Q100-009 AEC-Q003	Electrical Distributions: (Test @ Rm/Hot/Cold) (where applicable, Cpk > 1.67)	3	30	90	of	
FG	E6	AEC-Q100-007	Fault Grading:	-	-	-	Fault Grade {Select Level}	
CHAR	E7	AEC-Q003	Characterization: (Test @ Rm/Hot/Cold)	-	-	-	{Select Data}	
EMC	E9	SAE J1752/3	Electromagnetic Compatibility (Radiated Emissions)	1	1	1		
SC	E10	AEC Q100-012	Short Circuit Characterization	3	10	30		
SER	E11	JESD89-1 JESD89-2 JESD89-3	Soft Error Rate	1	3	3		
LF	E12	AEC-Q005	Lead (Pb) Free: (see AEC-Q005)	-	-	-		

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TEST GROUP F – DEFECT SCREENING TESTS

PAT	F1	AEC-Q001	Process Average Testing: (see AEC-Q001)	All	All	All	Reject units outside Avg.	
SBA	F2	AEC-Q002	Statistical Bin/Yield Analysis: (see AEC-Q002)	All	All	All	Reject units outside criteria	

TEST GROUP G – CAVITY PACKAGE INTEGRITY TESTS (for Ceramic Package testing only)

MS	G1	JESD22 B104	Mechanical Shock: (Test @ Rm)	1	15	15	of	
VFV	G2	JESD22 B103	Variable Frequency Vibration: (Test @ Rm)	1	15	15	of	
CA	G3	MIL-STD-883 Method 2001	Constant Acceleration: (Test @ Rm)	1	15	15	of	
GFL	G4	MIL-STD-883 Method 1014	Gross and Fine Leak:	1	15	15	of	
DROP	G5	-----	Drop Test: (Test @ Rm) MEMS cavity parts only. Drop part on each of 6 axes once from a height of 1.2m onto a concrete surface.	1	5	5	of	
LT	G6	MIL-STD-883 Method 2004	Lid Torque:	1	5	5	of	
DS	G7	MIL-STD-883 Method 2019	Die Shear:	1	5	5	of	
IWV	G8	MIL-STD-883 Method 1018	Internal Water Vapor:	1	3	3	of	